Properties of movement - Exercise

Animal tracking 25/26

Giuliano Colosimo

University of Rome Tor Vergata

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Properties of movement

Properties of movement

- The following exercise is adapted from the R script developed and presented by Kami Safi, Martina Scacco & Anne Scharf at the AniMove 2024 conference
- The R code will allow you to look at a series of very useful functions in the move2 package that allow us to calculate the movement metrics we talked in Class 5 slides
- You will need to use a series of packages outlined in the next slide and download some data from here
- Once downloaded, place the data folder somewhere on your computer

R Libraries

- If you haven't done it yet, install the following packages and proceed with loading them in your R working session

```
library(move2)
library(dplyr)
library(lubridate)
library(units)
library(classInt)
library(ggplot2)
library(sf)
library(mapview)
```

Data exploration

```
stwd("/PATH/TO/DATA/FOLDER")
bats <- mt_read("./Parti-colored bat Safi Switzerland.csv")
buff <- mt_read("./Kruger African Buffalo,
                GPS tracking, South Africa.csv.gz")
Leo <- mt_read("./animtrack/data/Leo-65545.csv")
fishers <- mt_read(mt_example())
```

Temporal organization of trajectories

Temporal organization of trajectories

```r
### Number of locations
nrow(bats)
```

```
[1] 1740
```

```r
### Number of tracks
mt_n_tracks(bats)
```

```
[1] 17
```

```r
### Number of locations per track
head(table(mt_track_id(bats)))
```

```

191 239 263 321 338 360 
 97 152  78 104  89 131 
```

```r
timeLags <- mt_time_lags(bats)
head(timeLags)
```

```
Units: [s]
[1]  780  600  720 2400 1320 1320
```

- By default the previous function uses the most convenient unit but we can change it or set it from the beginning

```r
timeLags <- set_units(timeLags, hours)
timeLags <- mt_time_lags(bats, units = "hour")
head(round(timeLags, 4))
```

```
Units: [h]
[1] 0.2167 0.1667 0.2000 0.6667 0.3667 0.3667
```

- If you remember from the slides of Class 5, timelag, distance, speed, and direction are properties of the segment (2 locations, not one)
- Therefore the number of valid values will be nrow() - 1 for each track
- Here we have 17 tracks and therefore 17 NAs

```r
mt_n_tracks(bats)
```

```
[1] 17
```

```r
table(is.na(timeLags))
```

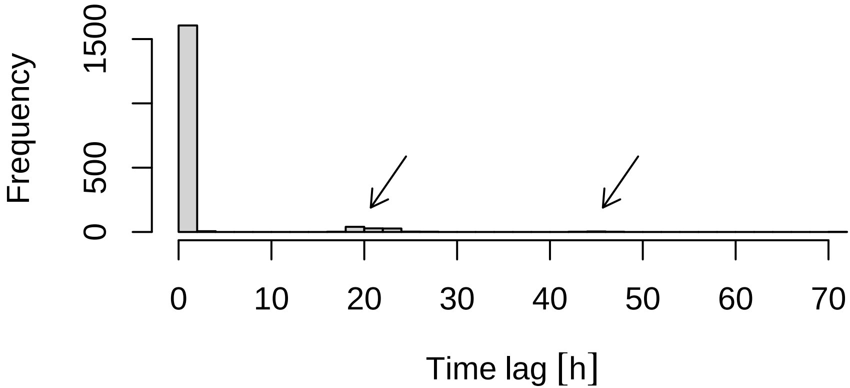
```
FALSE  TRUE 
 1723    17 
```

```r
tail(round(timeLags, 4))
```

```
Units: [h]
[1] 0.1333 0.4000 0.1333 0.1667 0.1333     NA
```

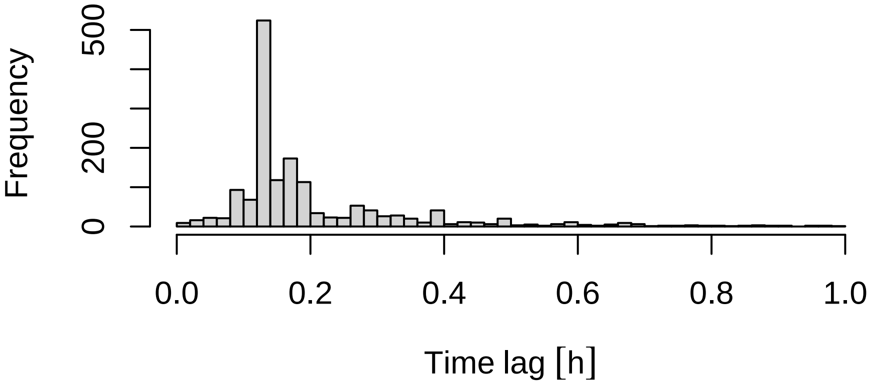
Distribution of time lags

```r
hist(timeLags, breaks=50, main=NA, xlab="Time lag")
arrows(24.5,587.5,20.7,189.7, length=0.1)
arrows(49.5,587.5,45.7,189.7, length=0.1)
```



Distribution of timelags shorter than 1h

```r
tl_short <- timeLags[timeLags < set_units(1, "hour")]
hist(tl_short, breaks=50, main=NA, xlab="Time lag")
```



Number of locations in time

```r
ts <- mt_time(bats)
head(ts, 10)
```

```
 [1] "2003-06-18 22:12:00 UTC" "2003-06-18 22:25:00 UTC"
 [3] "2003-06-18 22:35:00 UTC" "2003-06-18 22:47:00 UTC"
 [5] "2003-06-18 23:27:00 UTC" "2003-06-18 23:49:00 UTC"
 [7] "2003-06-19 00:11:00 UTC" "2003-06-19 00:23:00 UTC"
 [9] "2003-06-19 00:35:00 UTC" "2003-06-19 00:48:00 UTC"
```

- Make always sure you are working with the appropriate time zone, and, if not, transform time to local time zone of the study for better interpretation

- It is possible to check out the time zone of your system using:

```r
Sys.timezone()
```

```
[1] "Europe/Rome"
```

- You can check the available time zones using:

```r
# There are 497 different time zones,
# but the code here shows only results of the first 2
head(OlsonNames(), 2)
```

```
[1] "Africa/Abidjan" "Africa/Accra"
```

```r
tsLocal <- lubridate::with_tz(ts, tzone="Europe/Zurich")
head(tsLocal, 10)
```

```
 [1] "2003-06-19 00:12:00 CEST" "2003-06-19 00:25:00 CEST"
 [3] "2003-06-19 00:35:00 CEST" "2003-06-19 00:47:00 CEST"
 [5] "2003-06-19 01:27:00 CEST" "2003-06-19 01:49:00 CEST"
 [7] "2003-06-19 02:11:00 CEST" "2003-06-19 02:23:00 CEST"
 [9] "2003-06-19 02:35:00 CEST" "2003-06-19 02:48:00 CEST"
```

```r
# N. of location per hour (geometry gets inherited)
bats %>% group_by(hour(tsLocal)) %>%
  summarize(n()) %>% head(2)
```

```
Simple feature collection with 2 features and 2 fields
Geometry type: MULTIPOINT
Dimension:     XY
Bounding box:  xmin: 7.026008 ymin: 46.9032 xmax: 8.674103 yma
Geodetic CRS:  WGS 84
# A tibble: 2 x 3
  `hour(tsLocal)` `n()`
            <int> <int>
1               0   427 ((7.026008 46.92228), (7.044309 46.929
2               1   427 ((7.050854 46.92298), (7.058004 46.923
```

```r
# N. of locations per month and hour,
# rename columns and drop geometry
bats %>% group_by(Month = month(tsLocal),
                  Hour = hour(tsLocal)) %>%
  summarize(N.locations = n()) %>%
  sf::st_drop_geometry() %>% head(5)
```

```
# A tibble: 5 x 3
# Groups:   Month [1]
  Month  Hour N.locations
  <dbl> <int>       <int>
1     5     0         122
2     5     1         117
3     5     2          68
4     5     3          39
5     5     4          39
```

Spatial organization of trajectories

Distance between locations

- We can now start to look at the spatial organization of our data

```r
# if units is not set, it will use the most convenient
dist <- mt_distance(bats, units = "m")
summary(dist)
```

```
   Min. 1st Qu.   Median     Mean 3rd Qu.     Max.     NA's
    0.0   867.2   1729.6   2203.8  3022.1  15235.5       17
```

```r
hist(dist, main = "")
```

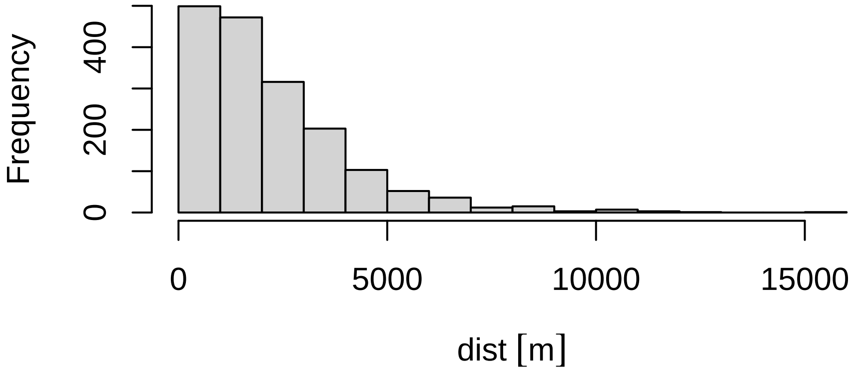


Figure 1: Distribution of distances (or step lengths)

Speed between locations

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```r
# if units is not set, it will use the most convenient
speeds <- mt_speed(bats, units = "m/s")
summary(speeds)
```

```
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
   0.0000    0.9057    2.3074    5.2485    4.7227 1096.6536 
```

```
hist(drop_units(speeds), breaks="FD", main = "")
```

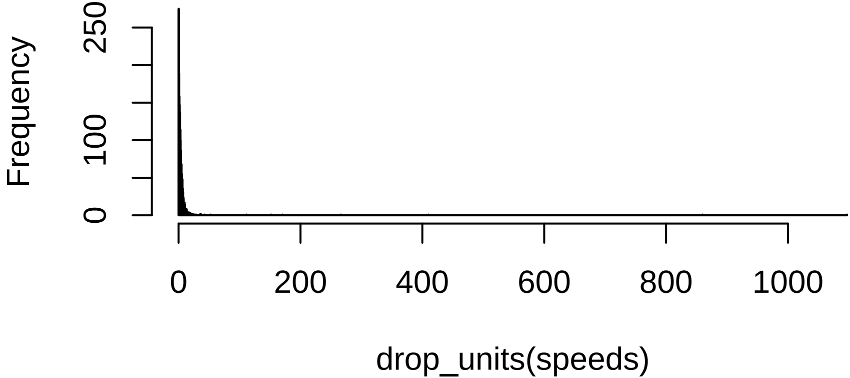


Figure 2: Distribution of speed between relocations

```r
# remember to set the unit before the operation
speedsRealistic <- speeds[speeds < set_units(20, m/s)]
hist(drop_units(speedsRealistic), main = "",
     xlab = "Speed [m/s]", breaks = "FD")
```

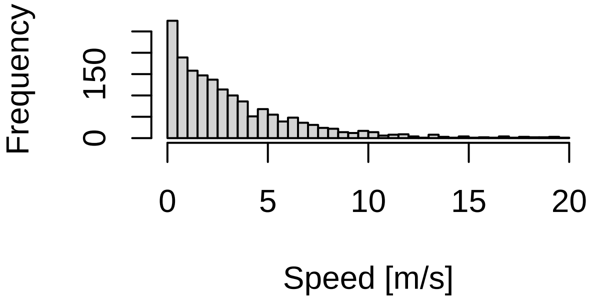


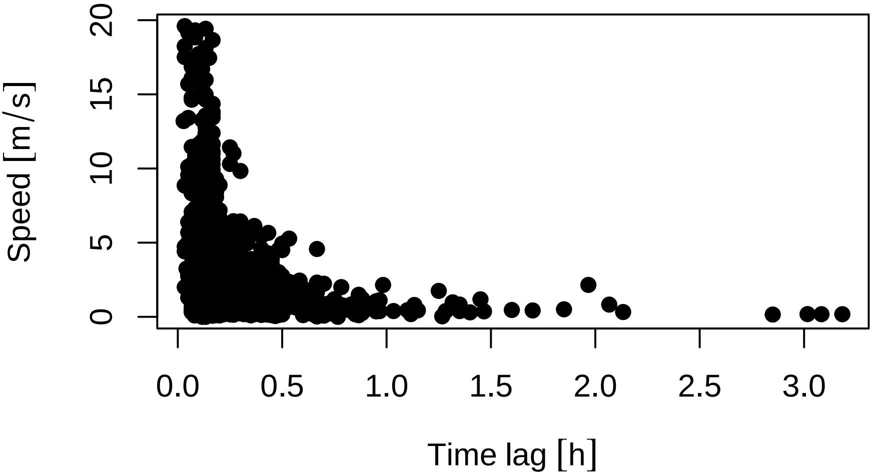
Figure 3: We can focus on the realistic speed (e.g. < 20m/s). This is the common shape for speeds. Animals mostly move slowly, sometimes they move fast

Speed VS Time lags

```
speedVsTimeLag <- data.frame(timeLag = timeLags,
                             speeds = speeds)

speedVsTimeLag <- speedVsTimeLag[speedVsTimeLag$timeLag
                                 < set_units(10, hour) &
                                   speedVsTimeLag$speeds <
                                   set_units(20, m/s),]
```

```r
plot(speedVsTimeLag$timeLag, speedVsTimeLag$speeds,
     xlab='Time lag', ylab='Speed', pch=19)
```



- We can make a plot to highlight segments with high and low speed

```
head(unique(mt_track_id(bats)))
```

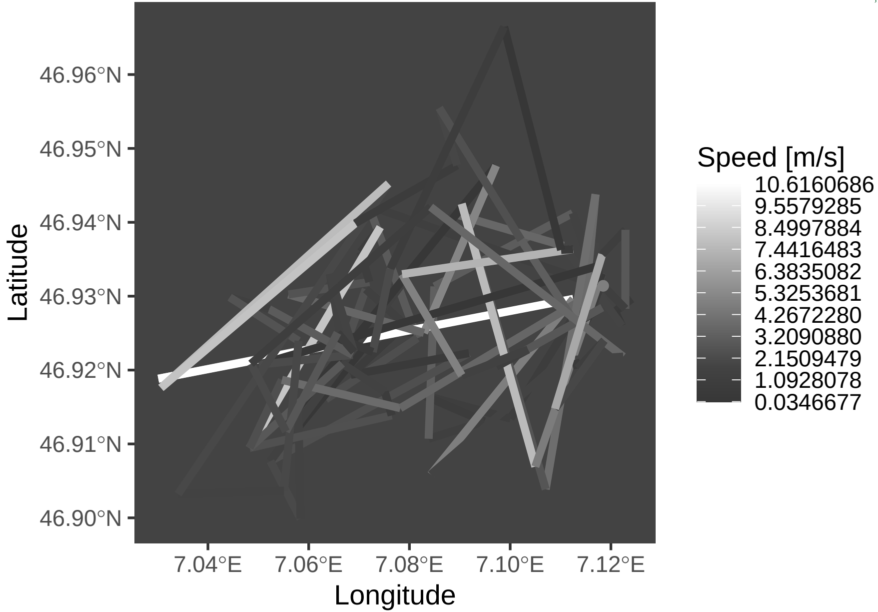
```
[1] 191 239 263 321 338 360
17 Levels: 191 239 263 321 338 360 553 021 042 127 146 168 289
```

```
# select bat 191
bat191 <- filter_track_data(bats, .track_id = "191")

# store speed
v <- set_units(mt_speed(bat191), m/s)

# find 9 colors for 10 breaks in the speed vector
myBreaks <- classIntervals(v[!is.na(v)], n=10,
                           style="equal")$brks
```

```
ggplot() +
  geom_sf(data = mt_segments(bat191),
          aes(color = drop_units(v)), linewidth = 1.5) +
  xlab("Longitude") + ylab("Latitude") +
  scale_color_gradient2(low = "black",
                  mid = "#434343", high = "white",
                  midpoint = drop_units(median(v, na.rm=T)),
                  breaks = drop_units(myBreaks),
                  name = "Speed [m/s]") +
  theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        panel.background = element_rect(fill = '#434343'))
```



Speed [m/s]
10.6160686
9.5579285
8.4997884
7.4416483
6.3835082
5.3253681
4.2672280
3.2090880
2.1509479
1.0928078
0.0346677
Latitude
46.96°N
46.95°N
46.94°N
46.93°N
46.92°N
46.91°N
46.90°N
Longitude
7.04°E
7.06°E
7.08°E
7.10°E
7.12°E

Azimuth

- Note: the words heading or bearing are mostly referred to the direction of body axis
- hen analysing tracking data we are not observing the animal but only its movement, so we can only know the direction of movement (not body orientation)

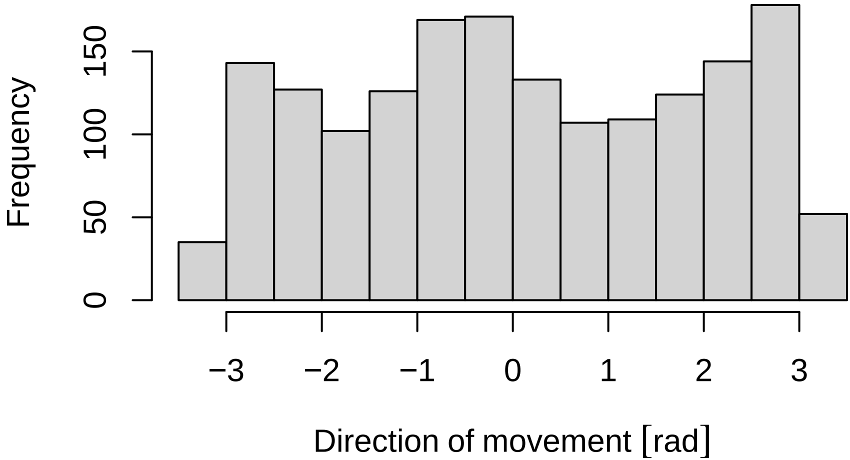
```
# Angles in radians relative to the N
direction <- mt_azimuth(bats)
head(round(direction, 4))
```

```
Units: [rad]
[1] -2.9576 -0.3009  0.2445 -2.5857 -1.3090  1.9759
```

```
summary(round(direction, 4))
```

```
    Min.  1st Qu.   Median     Mean  3rd Qu.     Max.     NA's
-3.14080 -1.36970 -0.03265  0.08658  1.80015  3.13980       20
```

```r
hist(direction,  breaks = 18,
     xlab="Direction of movement", main = NA)
```



Turning angles

- Turning angles are a property of 2 segments (3 locations)

```
turnAngles <- mt_turnangle(bats)
# angles in radians relative to the previous step
# with 2 NAs per track (first and last value of each track)
head(round(turnAngles[bats$`individual-local-identifier`
                      == 191],
      4))
```

```
Units: [rad]
[1]      NA  2.6567  0.5454 -2.8302  1.2767 -2.9983
```

```
head(summary(turnAngles))
```

```
       Min.     1st Qu.      Median        Mean     3rd Qu.
-3.14157433 -2.18668315  0.11312795  0.06951454  2.33746163  3
```

```r
hist(turnAngles, breaks = 18, xlab="Turning Angle",
     main = NA)
```

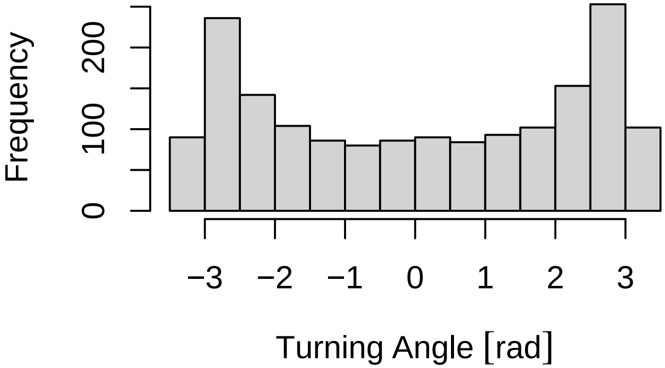


Figure 4: This shape can indicate movement along a linear structure

- Let's look at turning angle in a different animal, we use buffaloes

```r
turnAnglesBuf <- mt_turnangle(buff)
hist(turnAnglesBuf, main = "")
```